

CS 5990 - Assignment #2

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1. Adaptive Moving Average

Deviations:

$$|x_1 - \text{mean}(x_1, x_2, x_3)| = |20 - 21.67| = 1.67$$

$$|x_2 - \text{mean}(x_1, x_2, x_3)| = |18 - 21.67| = 3.67$$

$$|x_3 - \text{mean}(x_2, x_3, x_4)| = |27 - 22.67| = 4.33$$

$$|x_4 - \text{mean}(x_3, x_4, x_5)| = |23 - 24.67| = 1.67$$

$$|x_5 - \text{mean}(x_4, x_5, x_6)| = |24 - 21| = 3$$

$$|x_6 - \text{mean}(x_5, x_6, x_7)| = |16 - 22| = 6$$

$$|x_7 - \text{mean}(x_5, x_6, x_7)| = |26 - 22| = 4$$

S _t Formula	Original Value	Window	Mean	Std	Deviation	Threshold (k * std)	Smoothed?	New Value
$\bar{x}_1 = \frac{20 + 18 + 27}{3}$	20	20, 18, 27	21.67	4.73	1.67	4.25	1.67 < 4.25 NO	20
$\bar{x}_2 = \frac{20 + 18 + 27}{3}$	18	20, 18, 27	21.67	4.73	3.67	4.25	3.67 < 4.25 NO	18
$\bar{x}_3 = \frac{18 + 27 + 23}{3}$	27	18, 27, 23	22.67	4.51	4.33	4.06	4.33 > 4.06 YES	22.67
$\bar{x}_4 = \frac{27 + 23 + 24}{3}$	23	27, 23, 24	24.67	2.08	1.67	1.87	1.67 < 1.87 NO	23
$\bar{x}_5 = \frac{23 + 24 + 16}{3}$	24	23, 24, 16	21	4.36	3	3.92	3 < 3.92 NO	24
$\bar{x}_6 = \frac{24 + 16 + 26}{3}$	16	24, 16, 26	22	5.29	6	4.76	6 > 4.76 YES	22
$\bar{x}_7 = \frac{24 + 16 + 26}{3}$	26	24, 16, 26	22	5.29	4	4.76	4 < 4.76 NO	26

2. Aggregation

- Total Price = Price x Quantity

Date	Item	Category	Location	Total Price
9/5/2004	Watch	Electronics	Chicago	250
9/6/2004	Jacket	Apparel	New York	300
9/7/2004	Microwave	Home Appliances	Los Angeles	200
9/8/2004	Lego Set	Toys	San Francisco	200
9/9/2004	Textbook	Books	Chicago	300
9/10/2004	Mouse	Electronics	New York	225
9/11/2004	Toaster	Home Appliances	Los Angeles	150
9/12/2004	Doll	Toys	San Francisco	275

- Std of Price:
 $\sigma = 55.55$
- Std of Total Price:
 $\sigma = 50$

The reduction in standard deviation implies that aggregation provides more stable data with less variability.

- By Category

Category	Total Price
Electronics	475
Apparel	300
Home Appliances	350
Toys	475
Books	300

$$\sigma = 89.09$$

By Location

Location	Total Price
Chicago	550
New York	525
Los Angeles	350
San Francisco	475

$$\sigma = 88.98$$

Since the standard deviation of aggregation by location is lower than aggregation by category, we want to choose the aggregation by location as it produces more stable data. At the same time, aggregation by location also reduces the number of objects more than by category.

3. SMOTE

- $x = (102,200,150)$
 Nearest neighbor: $x_{nn} = (106,120,172)$
 $x_{new} = (102,200,150) + 0.5((106,120,172) - (102,200,150))$
 $x_{new} = (102,200,150) + 0.5(4, -80, 22)$
 $x_{new} = (102,200,150) + (2, -40, 11)$
 $x_{new} = (104,160,161)$
- $x = (188,159,113)$
 Nearest neighbor: $x_{nn} = (138,117,103)$
 $x_{new} = (188,159,113) + 0.5((138,117,103) - (188,159,113))$
 $x_{new} = (188,159,113) + 0.5(-50, -42, -10)$
 $x_{new} = (188,159,113) + (-25, -21, -5)$
 $x_{new} = (163,138,108)$

4. PCA

- Standardization
 $x_1 = 2, 5, 7, 10$
 $x_2 = 8, 9, 11, 6$
 $\mu_1 = 6, \sigma_1 = 3.37$
 $\mu_2 = 8.5, \sigma_2 = 2.08$
 $x_{11} = \frac{2 - 6}{3.37} = -1.187, \quad x_{12} = \frac{5 - 6}{3.37} = -0.297$
 $x_{13} = \frac{7 - 6}{3.37} = 0.297, \quad x_{14} = \frac{10 - 6}{3.37} = 1.187$

$$x_{21} = \frac{8 - 8.5}{2.08} = -0.24, \quad x_{22} = \frac{9 - 8.5}{2.08} = 0.24$$

$$x_{23} = \frac{11 - 8.5}{2.08} = 1.2, \quad x_{24} = \frac{6 - 8.5}{2.08} = -1.2$$

$$x_1 = [-1.187, -0.297, 0.297, 1.187]$$

$$x_2 = [-0.24, 0.24, 1.2, -1.2]$$

- Covariance Matrix

$$\begin{aligned} Cov(x_1, x_2) &= \frac{(-1.187 * -0.24) + (-0.297 * 0.24) + (0.297 * 1.2) + (1.187 * -1.2)}{3} \\ &= -0.285 \end{aligned}$$

$$\begin{bmatrix} Var(x_1) & Cov(x_1, x_2) \\ Cov(x_2, x_1) & Var(x_2) \end{bmatrix} = \begin{bmatrix} 1 & -0.285 \\ -0.285 & 1 \end{bmatrix}$$

- Eigenvectors/ Eigenvalues of covariance matrix

$$\det \left(\begin{bmatrix} 1 & -0.285 \\ -0.285 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = 0$$

$$\det \left(\begin{bmatrix} 1 - \lambda & -0.285 \\ -0.285 & 1 - \lambda \end{bmatrix} \right) = 0$$

$$(1 - \lambda)(1 - \lambda) - (-0.285 * -0.285) = 0$$

$$\lambda^2 - 2\lambda + 0.919 = 0$$

$$\lambda_{1,2} = \frac{2 \pm \sqrt{4 - 4(0.919)}}{2}$$

$$\lambda_1 = 1.285$$

$$(A - \lambda_1 I)X = 0$$

$$\begin{bmatrix} 1 - 1.285 & -0.285 \\ -0.285 & 1 - 1.285 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -0.285 & -0.285 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -0.285x_1 - 0.285x_2 \end{bmatrix} = 0$$

$$x_1 = -x_2$$

$$v_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0.707 \\ -0.707 \end{bmatrix}$$

$$\lambda_2 = 0.715$$

$$(A - \lambda_2 I)X = 0$$

$$\begin{bmatrix} 1 - 0.715 & -0.285 \\ -0.285 & 1 - 0.715 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 0.285 & -0.285 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 0.285x_1 - 0.285x_2 \end{bmatrix} = 0$$

$$x_1 = x_2$$

$$v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$$

- Selecting Principal Components

Since $\lambda_1 > \lambda_2$, pick the larger one (λ_1)

- Recast the data

$$X = \begin{bmatrix} -1.187 & -0.24 \\ -0.297 & 0.24 \\ 0.297 & 1.2 \\ 1.187 & -1.2 \end{bmatrix} v_1 = \begin{bmatrix} 0.707 \\ -0.707 \end{bmatrix}$$

$$Z = \begin{bmatrix} -0.67 \\ -0.38 \\ -0.64 \\ 1.69 \end{bmatrix}$$

5. ANOVA

- Prepare the dataset

Age (sample)	Cholesterol (X_1)	Blood pressure (X_2)	Heart disease (Y)
45	220	130	Yes
50	250	140	Yes
38	190	125	No
55	270	145	Yes
42	200	128	No

- Compute means

Mean of X_1 :

- Class 0 (yes): $\frac{220+250+270}{3} = 246.67$
- Class 1 (no): $\frac{190+200}{2} = 195$
- Total mean: $\frac{220+250+190+270+200}{5} = 226$

Mean of X_2 :

- Class 0 (yes): $\frac{130+140+145}{3} = 138.33$
- Class 1 (no): $\frac{125+128}{2} = 126.5$
- Total mean: $\frac{130+140+125+145+128}{5} = 133.6$

- Compute variance between groups

- $SS_{B1} = (3 * (246.67 - 226)^2) + (2 * (195 - 226)^2) = 3203.75$
- $SS_{B2} = (3 * (138.33 - 133.6)^2) + (2 * (126.5 - 133.6)^2) = 167.94$

- Compute variance within groups

- $SS_{w1} = (220 - 246.67)^2 + (250 - 246.67)^2 + (190 - 195)^2 + (270 - 246.67)^2 + (200 - 195)^2 = 1316.67$
- $SS_{w2} = (130 - 138.33)^2 + (140 - 138.33)^2 + (125 - 126.5)^2 + (145 - 138.33)^2 + (128 - 126.5)^2 = 121.17$

- Compute F-scores

$$F_1 = \frac{3203.75/1}{1316.67/3} = 7.3$$

$$F_2 = \frac{167.94/1}{121.17/3} = 4.16$$

- Interpret the results

Since $F_1 > 5$, keep the feature (Cholesterol). On the other hand, since $F_2 < 5$, Blood pressure feature can be eliminated.

6. Github repo: <https://github.com/fidelisprasetyo/CS5990/tree/hw2>